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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one composition of morphisms between fibre products; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definition of a formal algebraic space, and locality of that condition along representable etale coverings. Same chapter, Definition definition-formal-algebraic-space at proof lines 3495-3497 and Lemma lemma-etale-covering-by-formal-algebraic-spaces at proof lines 3500-3503. Reduce the statement to the case where both spaces are affine.
- Affine formal algebraic spaces as filtered colimits of affine schemes with thickenings as transition maps. Same chapter, Definition definition-affine-formal-algebraic-space, invoked at proof lines 3507-3509. Supplies the presentations whose fibre products are taken.
- A filtered colimit of algebraic spaces glued by thickenings is a formal algebraic space. Same chapter, Lemma lemma-colimit-is-formal, invoked at proof lines 3514-3518. Gives the conclusion in the affine case.

Source reference. Original lemma 0AIU: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2973–2973. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJ1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3470–3470. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
    (a) each X_λ is affine,
    (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Definition 0.2. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i\in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.3. *Let S be a scheme. Let X, Y be formal algebraic spaces over S and let Z be a sheaf whose diagonal is representable by algebraic spaces. Let $X \to Z$ and $Y \to Z$ be maps of sheaves. Then $X \times_Z Y$ is a formal algebraic space.*

Proof. Choose $\{X_i \to X\}$ and $\{Y_j \to Y\}$ as in Definition 0.2. Then $\{X_i \times_Z Y_j \to X \times_Z Y\}$ is a family of maps which are representable by algebraic spaces and étale. Thus Lemma 0AJ1 tells us it suffices to show that $X \times_Z Y$ is a formal algebraic space when X and Y are affine formal algebraic spaces.

Assume X and Y are affine formal algebraic spaces. Write $X = \operatorname{colim} X_\lambda$ and $Y = \operatorname{colim} Y_\mu$ as in Definition 0.1. Then $X \times_Z Y = \operatorname{colim} X_\lambda \times_Z Y_\mu$. Each $X_\lambda \times_Z Y_\mu$ is an algebraic space. For $\lambda \leq \lambda'$ and $\mu \leq \mu'$ the morphism

$$X_\lambda \times_Z Y_\mu \to X_\lambda \times_Z Y_{\mu'} \to X_{\lambda'} \times_Z Y_{\mu'}$$

is a thickening as a composition of base changes of thickenings. Thus we conclude by applying Lemma 0AIU. □